

Nahin Intesher

Computer Science Student

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 Website |  LinkedIn |  GitHub

Profile

Final-trimester Computer Science and Engineering student interested in Human-Computer Interaction and assistive technologies integrated with Computer Vision. Seeking opportunities to build intelligent, human-centric systems for practical applications, with interests in academic teaching, research, collaboration, and further studies.

Education

B.Sc. in Computer Science & Engineering

2022–Present

United International University · Dhaka, Bangladesh

CGPA: 3.93/4.00 (through the 11th trimester)

Relevant coursework: Data Structures & Algorithms, Database Management Systems, Software Engineering, Computer Networks, Artificial Intelligence, Data Mining, Machine Learning, Digital Image Processing.

Higher Secondary Certificate (Science)

2018–2020

Dhaka Residential Model College · Dhaka, Bangladesh

GPA: 5.00/5.00

Secondary School Certificate

2016–2018

Shamshul Haque Khan School and College · Dhaka, Bangladesh

GPA: 5.00/5.00

Junior School Certificate (Science)

2015

Shamshul Haque Khan School and College · Dhaka, Bangladesh

GPA: 5.00/5.00

Research Interests

Human-Computer Interaction (HCI) • Computer Vision • Quantum Machine Learning

Teaching Experience

Undergraduate Teaching Assistant

2023–2026

United International University · Dhaka, Bangladesh

- Helped students understand academic problems and technical concepts through practical exercises.
- Supported students in Data Structures and Algorithms, Object-Oriented Programming, Structured Programming Language, and Introduction to Computer Systems.
- Conducted lab sessions, mentored students in programming and debugging, held consultations, and graded lab work and assessments with constructive technical feedback.

Owner & Academic Instructor

2024–Present

Diganta Coaching Center · Dhaka, Bangladesh

- Teach students in grades 6–10, explain topics in accessible ways, and encourage them to explore their interests and opportunities to contribute.

Research Experience

Bridging the Gap: Towards Building AI Awareness among Vulnerable Populations

Manuscript in preparation

Undergraduate Thesis

- Investigated awareness and perceptions of AI-generated synthetic media, including multimodal deepfakes, among older adults aged 55+ and younger participants aged 11–18.
- Used participatory design, interviews, and surveys to understand participants' perceptions of deepfake content; trained a model and developed a mobile AI-awareness tool informed by participant data.
- Examined the problem from Computer Vision, Human-Computer Interaction, and Social Computing perspectives.

AI-Assisted Parenting

2024

Qualitative Study (N = 5)

- Studied mental workload, emotional stressors, and trust in AI tools for child development and medical decisions.
- Performed thematic analysis and developed user personas and design guidelines for trustworthy parental AI assistants.

Reproducing and Evaluating Security Mechanisms for the Model Context Protocol (MCP)

Author

- Re-implemented four MCP security mechanisms (SAMOS, AgentBound, MCPShield, and Breaking the Protocol) on 299 MCP servers.
- Conducted controlled evaluations of information-flow control, capability restrictions, and metadata misalignment detection, reporting a 78.38% detection rate for metadata misalignment.
- Analyzed security vulnerabilities in LLM-agent tool integration and compared reproduced results with reported metrics.

NutriSight: An AI-Powered Multi-Food Recognition and Personalized Nutrition Insight System

Primary Author

- Developed a multi-stage vision pipeline using YOLO26n-Seg, ViT-LSTM, and ViT-Small (Nutrition5k) for multi-food detection and calorie estimation.
- Achieved an F1-score of 0.9007 for custom food recognition and a test MAE of 18.05 for meal-level nutrition estimation.
- Built an Expo mobile app with a FastAPI backend to deliver personalized dietary insights based on user profiles.

Robustness-Efficiency Trade-offs of INT8 Quantization on Edge Devices

2026

Primary Author

- Evaluated clean accuracy and corruption robustness on CIFAR-10-C for FP32, INT8 PTQ, and INT8 QAT models on edge hardware.
- Proposed a degradation-aware PTQ calibration strategy (INT8-RPTQ) to improve noise robustness without full QAT retraining.
- Analyzed hardware performance trade-offs in latency, peak RAM, power consumption, and energy per inference.

Projects

Deepfake Detection System (DeepShield) [\[link\]](#)

06/2025–10/2026

Thesis Project

- Developed a video deepfake detection system combining a Swin Transformer and DCT backbone with a Temporal Transformer.
- **Datasets:** Celeb-DF-v2, FaceForensics++_C23.
- **Technologies:** Python, PyTorch, OpenCV.

Jiggasha [\[link\]](#)

04/2025–06/2025

Gamified Educational Platform

- Developed an online platform to support exam preparation through a fun, competitive learning experience.
- Implemented a battle-royale-style exam system with live exams and participant rankings.
- **Technologies:** Next.js, Node.js, PostgreSQL.

DiscoverYou [\[link\]](#)

06/2024–08/2024

Talent Discovery & Development Platform

- Built a platform for discovering, developing, and showcasing talents through contests, courses, communities, and expert guidance.
- Connected users with job opportunities and a marketplace to monetize their creative work.
- **Technologies:** React, Node.js, MySQL.

Diganta Coaching Center [\[link\]](#)

05/2025–Present

Coaching Center Website

- Developed a website for my coaching center to facilitate communication between students and teachers.
- **Technology:** Next.js.

CV Banao Mobile App [\[link\]](#)

2026

CV Building App

- Built a dedicated CV builder for academic, research, PhD and higher-study, and industry applications.
- **Technologies:** React Native, TypeScript.

Mini Game Master [\[link\]](#)

08/2024

Multiplayer Game

- Developed a multiplayer game offering a collection of classic mini-games.

Awards & Scholarships

Academic Scholarship

Awarded 100% tuition scholarships in nine trimesters and 50% tuition scholarships in three additional trimesters.

UIU CSE Project Show

Summer 2024–Summer 2025

United International University

- **Champion**, Database Management System — DiscoverYou.
- **Champion**, System Analysis and Design — Jiggasha.
- **2nd Runner-up**, Software Engineering — DiscoverYou Mobile App.
- **3rd Runner-up**, Advanced Object-Oriented Programming — MiniGameMaster.

UIU Juniors Programming Contest

2024

1st Runner-up

Competitive Programming

Codeforces	Nahinintesher	270 problems solved
LeetCode	Nahinintesher	41 problems solved

Skills & Methods

Programming	Python, C, C++, JavaScript, TypeScript
AI / Machine Learning	PyTorch, OpenCV
Web Development	Next.js, React, Node.js, Express.js, React Native
Databases	PostgreSQL, MySQL
Tools	Git, GitHub, Linux, Adobe Illustrator

References

Dr. Novia Nurain

Assistant Professor, Department of Computer Science & Engineering
Bangladesh University of Engineering and Technology (BUET)
Contact details available on request.